

FR-018 M4 — Email Job Alert Acquisition

FR018 — `sources/optional/m4.md`

Masterclass source package snapshot

Functional Requirement	FR-018
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Accepted** — engineering evaluation (live validation complete)
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via `python scripts/build_masterclass_package.py`. Repository documentation remains the source of truth.

Contents

Original M4 design (superseded for content)

FR-018 M4 — Email Job Alert Acquisition

Status: **Accepted** — engineering evaluation (live validation complete)

Date: 2026-08-07

Freeze: [FR-018 acceptance](#)

Phase: Horizon 1B (Opportunity Acquisition Framework)

Preceding: [M0–M3](#); architecture spike (Hybrid / Option C accepted)

ADR: [ADR-010](#)

1. Executive summary

M4 adds the second production acquisition channel under FR-018:

owner-saved job-alert .eml → **parse** → **Discovery Ingress** → **email acquire (with optional URL enrichment)** → **frozen Horizon 1A.**

Capability	Status
SEEK job-alert digests	Supported
LinkedIn job-alert digests	Supported — parse + URL enrich for full ads
Indeed job-alert digests	Supported (parse; enrich when URL fetch succeeds)
Unsupported senders / no job URLs	Fail closed
Recruiter CRM / outreach mail	Out of scope
Horizon 1A / ingress thinness	Unchanged

Verdict: **ACCEPTED** (owner live validation). Canonical freeze record:

[fr018_opportunity_discovery_acquisition.md](#).

2. Architecture (final accepted)

```
Job alert .eml (owner-saved)
  → parse_job_alert_email (MIME + allow-listed sender)
  → N × OpportunitySource (path.eml#job=i)
  → ThinDiscoveryIngress
  → EmailAcquisitionAdapter
    → card text from alert
    → if card-only / thin: UrlAcquisitionAdapter(job_url) [fail-soft]
  → AcquisitionResult(source_kind=email, raw_content=enriched JD when available)
  → ApplicationWorkflowRunner.start
  → Opportunity (FR-009)
```

No second orchestrator, no second Opportunity store, no IMAP client in M4 (owner supplies `.eml` files — mailbox auth deferred).

Original M4 design (superseded for content)

Email → parse card → Horizon 1A

Live LinkedIn validation proved alert bodies are **discovery cards** (~500 chars), not authoritative JDs. Extending email parsing was rejected in favour of **reusing the canonical URL acquisition path**.

3. Email acquisition design

Component	Location
MIME parse / sender classify / job URL extract	<code>discovery/email_parse.py</code>
<code>EmailAcquisitionAdapter</code> (+ optional URL enrich)	<code>discovery/email_adapter.py</code>
URL body fetch/extract	<code>discovery/url_adapter.py</code> (reused)
Provenance	<code>assert_email_acquisition_provenance</code>
Contracts	<code>DiscoverySourceKind</code> includes <code>"email"</code> ; locator <code>path.eml#job=N</code>
CLI	<code>cic opportunity discover-email</code>
Fixtures	<code>tests/fixtures/discovery/*_job_alert.eml</code>

One posting per adapter call (ADR-010). Digests expand at CLI/source-builder into N sources — Protocol not widened to `acquire_many`.

Content strategy: Prefer alert body when it already carries JD signals; when the alert is card-only (common for LinkedIn), **enrich via** `UrlAcquisitionAdapter` on the extracted job URL (fail-soft). Offline fixtures skip enrichment. Job URL remains required for FR-009 identity facets. No Playwright.

4. Supported email sources

Source	From signal	Extract	Notes
SEEK	seek.com.au / jobmail	/job/<id> links	Digests → multiple jobs
LinkedIn	linkedin.com job alerts	/jobs/view/<id> and / comm/jobs/view/<id>/	Plaintext Title / Company / View job: cards; enrich via URL adapter
Indeed	indeed.com alerts	jk= links	Enrich when HTTP extract succeeds
Other	—	—	unsupported_source
SEEK with no job URLs	seek sender	—	unsupported_source

Unsupported (documented): recruiter conversation email, newsletters, calendar invites, HTML-only marketing without board job URLs, IMAP sync.

5. Provenance

Field	Email rule
source_kind	"email" (unchanged after URL enrich)
source_url	Job URL from alert (required)
source_identifier	{Message-ID}#job={index}
acquired_at	Set
Warnings	email_message_id, email_from, email_path, platform; enriched_from_job_url or job_url_enrichment_failed

URL assert unchanged. Email assert is parallel and fail-closed.

6. Idempotency

Reuses FR-009 definite identity only (identity_probe_from_url on extracted job URL; post-acquire posting probe). Duplicate alert for the same SEEK/LinkedIn/Indeed job → SKIPPED_ALREADY_REPRESENTED . No separate email dedupe store.

Live: second discover-email without --force → skipped=6 .

7. Testing

Suite	Result
tests/unit/discovery/ (incl. enrich + live .eml regression)	Passed
Offline fixture path	Marker present; no live URL enrich
Enrich fail-soft	URL 403/blocked keeps email card

Coverage: supported parse, /comm/jobs/view/, unsupported sender, empty SEEK alert, adapter provenance, digest expansion, ingress acquire + duplicate skip, URL enrich success/fail-soft, URL regression.

8. Manual / live validation

Input	Result
Owner linkedin_alert.eml + --force	acquired=6 failed=0 ; full Horizon 1A artefacts
Same file without --force	skipped=6 (definite identity)
Fixture SEEK / LinkedIn / Indeed	Unit + offline CLI smoke

9. Lessons learned (Academy)

1. Unit tests ≠ product readiness for external integrations.
 2. Email alerts discover; they do not authorise the job description.
 3. Prefer reuse of URL acquisition over email-parse escalation.
 4. Live validation simplified architecture.
 5. Live validation is mandatory for board/email integrations.
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10. Documentation

Acceptance freeze: [fr018_opportunity_discovery_acquisition.md](#); changelog § 1.125; ADR-010 close-out note; Academy [masterclass/FR018/](#).

11. Out of scope (unchanged)

IMAP mailbox sync; Playwright; Easy Apply / authenticated LinkedIn apply; recruiter CRM; scheduled discovery; new boards beyond SEEK/LinkedIn/Indeed attempt.